



purely electronic signals to dramatically alter the functioning of a human brain. In the case of Parkinson's disease or epilepsy, the result is to override or counteract the nature of the disease, restoring the individual to a semblance of normalcy. In this sense, the use of electronic signals can be seen as a viable alternative, and in many cases, an improvement on the more traditional chemical alternative.

Electronic signals can be used to alter the workings of the human brain in a variety of ways. For instance, we can consider the possibilities of memory download—giving individuals memories, which hitherto, they have never experienced.

This can even be extended to education. Want to play golf like Tiger Woods? Well, perhaps the mental ability can be downloaded, even if physically you might be a little overweight or out of condition. But potentially, even that could be dealt with. Why not try the patented 'Slimplant'—electronic signals used to control a brain's functioning regarding eating and drinking patterns and the amount of regular exercise?

Research on rats has shown that their planned movements can be controlled externally by the computer. When this is transferred to the human

mathematical and memory features of a computer directly linked with your brain, and being able to think in many dimensions.

Through my own implant self-experimentation research, I have already experienced an extra, ultrasonic sense. I have directly controlled a robot hand on a different continent by brain signals alone, and have communicated directly from one nervous system to another.

To me, the prospect of upgrading from the limited world of a human into an exciting part-human part-machine world—a 'Cyborg', in science fiction terms, is so exciting that I cannot wait. Although many others will feel like I do, some may feel quite differently.

For instance, if some humans have extra abilities, how will they treat those that remain mere humans? Nevertheless, just because some people do not want to be upgraded, why should they stop those who want to be upgraded? Clearly, we face tough ethical questions in the years to come.

One area where this is highlighted is the military domain. Implant technology opens many avenues to a whole new world of technological warfare. Not only can those involved have more improved intellectual capabilities, including extra

“With a nervous system that can be extended via a network, human soldiers can be safe at home, while some of their senses are remote, at the point of combat.”

brain it means that you will not be able to have that extra meal that you know you shouldn't; the implant will not let you! Of course, this could put McDonald's out of business!

All sorts of behavioural patterns could be controlled in the same way. For instance, one individual could potentially control some of the movements and decisions made by another. Not sure what your spouse is doing? Well, now you could be, via an implant.

Conversely, a person might want such signals to allow them to deal confidently with an interview, or make a presentation, or to do the right thing on a first date. The possibilities are endless. Any time you are not sure—rely on the implant.

At the same time, such possibilities raise enormous social and ethical questions. It is one thing to use this technology for therapeutic purposes in order to help someone who has, say, Parkinson's disease or paralysis—very few people would disagree with such usage.

However, it's a very different thing when it comes to controlling another individual's habits or movements. Even if technology allows it, the society on the whole may not be keen on the idea. Society is, indeed, a strange creature.

The ethical issues are further exacerbated when enhancement of an individual is contemplated. The ability to communicate by thought is one feature. However, brain implants offer the possibilities of: extra-sensory input (such as infrared, ultrasonics), extending your nervous system across the Internet (your brain can directly control technology in cyberspace), having the

senses, but also, humans themselves no longer need to be physically present in the battlefield.

With a nervous system that can be extended via a network, the human soldier, pilot or sailor can be safely ensconced at home, while some of their senses are remote, at the point of combat. Any thoughts of the individual on control or movement, as would be the case in firing a missile, can then be translated into immediate, direct commands in the combat zone.

From the home side's perspective, this has the distinct advantage of saving the lives of those in combat. For the opposition, however, it becomes more of a challenge. Destroying a vehicle or machine would be just that and only that. If it can be replaced, it will be.

Moreover, the attacking machine will need no intelligence, no computer, and no high technology on board. Effectively, it can almost become a 'throw-away', low-cost artificial soldier, with the actual personnel residing safely at home.

Space travel, too, would take on a distinctly different meaning. Why send astronauts into the extremely dangerous environment of outer space when they can stay at home and have their nervous system linked to a distant planet, by satellite?

In this way, we will soon be able to experience planets at the extremes of our solar system, which would, at present, appear to be impossible otherwise—simply because of the time required to safely transport humans across space for such a distance, never mind bringing them back again.

In all probability, we will develop space travellers that spring to life when they arrive at a